ECO 202: Formula and Concept Review

Classes 10–13 (material after Exam 2)

1. Random Variables and Distributions

For a discrete random variable,

$$p_X(x) = \mathbb{P}(X = x)$$

The cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \le x)$$

$$\mathbb{P}(a < X \le b) = F_X(b) - F_X(a)$$

Expectation

$$\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x)$$

$$\mathbb{E}[g(X)] = \sum_x g(x)\mathbb{P}(X = x)$$

$$\mathbb{E}[a + bX] = a + b\mathbb{E}[X]$$

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

Linearity of expectation does **not** require independence. For an indicator $H = \mathbf{1}\{A\}$,

$$\mathbb{E}[H] = \mathbb{P}(A)$$

2. Variance and Standard Deviation

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$$

$$\mathrm{Var}(a + bX) = b^2 \,\mathrm{Var}(X)$$

$$\mathrm{SD}(a + bX) = |b| \,\mathrm{SD}(X)$$

- Variance has squared units
- Standard deviation has the original units
- Adding a constant changes neither variance nor SD

3. Covariance and Correlation

For discrete variables, independence means

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y) \quad \text{for all } x, y$$

Under independence,

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\mathrm{SD}(X)\,\mathrm{SD}(Y)}$$

$$-1 \le \mathrm{Corr}(X, Y) \le 1$$

Covariance has product units; correlation is unitless. Neither alone establishes causality.

Variance of a sum

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y)$$

$$\mathrm{Var}(aX + bY) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y)$$

Independence versus zero covariance

If X and Y are independent and have finite variances,

$$X \perp Y \quad \Longrightarrow \quad \mathrm{Cov}(X, Y) = 0$$

But the converse is false:

$$\mathrm{Cov}(X, Y) = 0 \quad \not\Longrightarrow \quad X \perp Y$$

Zero covariance is nevertheless enough to make variances add:

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

Counterexample: Let U be equally likely to equal -1, 0, or 1, and let $V = U^2$. Then $\mathrm{Cov}(U, V) = 0$, but U and V are dependent because V is determined by U.

Review: Lecture 10, Section 5 and Practice Problem 5 give the counterexample. Precept 6, Problem 1(f), reviews how to test independence.

4. Bernoulli Variables and Counts

If $H \sim \mathrm{Bernoulli}(p)$,

$$\mathbb{P}(H = 1) = p, \qquad \mathbb{P}(H = 0) = 1 - p$$

$$\mathbb{E}[H] = p, \qquad \mathrm{Var}(H) = p(1 - p)$$

If $H_1, \ldots, H_n$ are independent $\mathrm{Bernoulli}(p)$ and

$$K = \sum_{i=1}^{n} H_i$$

then

$$K \sim \mathrm{Binomial}(n, p)$$

$$\mathbb{E}[K] = np, \qquad \mathrm{Var}(K) = np(1 - p)$$

$$\mathrm{SD}(K) = \sqrt{np(1 - p)}$$

Sample proportion

$$\widehat{p} = \frac{K}{n} = \frac{1}{n}\sum_{i=1}^{n} H_i$$

$$\mathbb{E}[\widehat{p}] = p$$

$$\mathrm{Var}(\widehat{p}) = \frac{p(1-p)}{n}$$

$$\mathrm{SE}(\widehat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

- p: fixed model probability
- $\widehat{p}$: random estimator before observing data
- Observed proportion: one realized estimate

5. Sample Means and Standard Errors

Suppose $X_1, \ldots, X_n$ are independent with common mean μ and variance σ^2.

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$\mathbb{E}[\overline{X}] = \mu$$

$$\mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}$$

$$\mathrm{SE}(\overline{X}) = \mathrm{SD}(\overline{X}) = \frac{\sigma}{\sqrt{n}}$$

If σ is unknown, the estimated standard error is

$$\widehat{\mathrm{SE}}(\overline{X}) = \frac{s}{\sqrt{n}}$$

Multiplying n by 4 divides the standard error by 2.

6. LLN and CLT

Law of Large Numbers

$$\overline{X}_n \xrightarrow{p} \mu$$

The unscaled sample average concentrates around μ.

Central Limit Theorem

Under the appropriate independence and finite-variance conditions,

$$\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$$

For a sufficiently large finite n,

$$\overline{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

Standardize with

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}}$$

- LLN: where the average concentrates
- CLT: approximate shape after centering and scaling
- The CLT approximation is not exact
- The observations themselves do not become Normal
- Large n does not repair selection, measurement, or dependence problems

7. Continuity Correction

Use a half-unit boundary when approximating an integer-valued count by a continuous Normal variable:

$$K \geq k \quad \longrightarrow \quad N \geq k - 0.5$$

$$K \leq k \quad \longrightarrow \quad N \leq k + 0.5$$

$$a \leq K \leq b \quad \longrightarrow \quad a - 0.5 \leq N \leq b + 0.5$$

For $K \sim \mathrm{Binomial}(n, p)$,

$$z = \frac{\text{continuous boundary} - np}{\sqrt{np(1-p)}}$$

Check expected successes and failures:

$$np \qquad \text{and} \qquad n(1-p)$$

8. Estimation, Bias, and MSE

Estimand	Fixed target θ
Estimator	Random rule T
Estimate	Realized numerical value t
Error	$T - \theta$
Sampling distribution	Distribution of T under repeated use of the stated mechanism

$$\mathrm{Bias}(T;\theta) = \mathbb{E}[T] - \theta$$

$$T \text{ is unbiased for } \theta \quad \Longleftrightarrow \quad \mathbb{E}[T] = \theta$$

$$\mathrm{MSE}(T;\theta) = \mathbb{E}[(T-\theta)^2]$$

$$\mathrm{MSE}(T;\theta) = \mathrm{Var}(T) + \mathrm{Bias}(T;\theta)^2$$

$$\mathrm{SE}(T) = \sqrt{\mathrm{Var}(T)}$$

- Unbiased does not mean correct in every sample
- Smaller bias does not automatically mean smaller MSE
- Smaller variance does not automatically mean smaller MSE
- Estimator bias is different from selection, nonresponse, or measurement bias